



**GOKUL
GLOBAL
UNIVERSITY**
SIDHPUR, GUJARAT



C++ Practical Manual

1.Reverse a Number

```
#include <iostream>
using namespace std;

int main() {
    int num, reverse=0, digit;

    cout<<"Enter a number: "; cin >>
    num;

    while (num != 0) {
        digit=num%10;
        reverse=reverse*10+digit; num =
        num / 10;
    }

    cout<<"Reverse number="<<reverse<<endl; return
    0;
}
```

2. Factorial of a Number

```
#include <iostream>
using namespace std;

int main() {
    int n;
    long long factorial=1;

    cout<<"Enter a number:"; cin >>
    n;

    if(n<0) {
        cout<<"Factorial of negative number is not possible."<<endl;
    }
    else {
        for(int i=1; i<=n; i++) {
            factorial=factorial* i;
        }

        cout<<"Factorial="<<factorial<<endl;
    }

    return 0;
}
```

3.Positive,Negative or Zero

```
#include <iostream>
using namespace std;

int main() {
    int num;

    cout<<"Enter a number:"; cin >>
    num;

    if (num>0) {
        cout<<"Number is Positive."<< endl;
    }
    elseif (num<0) {
        cout<<"Number is Negative."<< endl;
    }
    else {
        cout<<"Number is Zero."<<endl;
    }

    return 0;
}
```

4.Fibonacci Series

```
#include <iostream>
using namespace std;

int main() {
    int n;
    int first=0, second=1, next;

    cout<<"Enter number of terms:"; cin >>
    n;

    cout<<"Fibonacci Series:";

    for (int i=1; i<=n; i++) { cout <<
        first <<" ";

        next=first+second; first
        = second;
        second=next;
    }

    cout<<endl; return
    0;
}
```

1.Star Triangle

```
#include <iostream>
using namespace std;

int main() {
    int n;

    cout<<"Enter number of rows:"; cin >>
    n;

    for (int i=1; i<=n; i++) {
        for (int j=1; j<=i; j++) {
            cout<<"*";
        }
        cout<<endl;
    }

    return 0;
}
```

Input:

5

2. Reverse Star Triangle

```
#include <iostream>
using namespace std;

int main() {
    int n;

    cout<<"Enter number of rows:"; cin >>
    n;

    for (int i=n; i>=1; i--) {
        for (int j=1; j<=i; j++) {
            cout<<"*";
        }
        cout<<endl;
    }

    return 0;
}
```

Input:

5

3. Number Triangle

```
#include <iostream>
using namespace std;

int main() {
    int n;

    cout << "Enter number of rows: "; cin >>
    n;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            cout << j;
        }
        cout << endl;
    }

    return 0;
}
```

Input:

5

4. Pyramid Pattern

```
#include <iostream>
using namespace std;

int main() {
    int n;

    cout << "Enter number of rows: "; cin >>
    n;

    for (int i = 1; i <= n; i++) {

        //Print spaces
        for (int j = 1; j <= n - i; j++) { cout << " ";
        }

        //Print stars
        for (int j = 1; j <= (2 * i - 1); j++) { cout << "* ";
        }

        cout << endl;
    }

    return 0;
}
```

1. LargestofTwoNumbers

```
#include <iostream>
using namespace std;

int main()
{
    int a, b;

    cout << "Enter first number: "; cin
    >> a;

    cout << "Enter second number: "; cin
    >> b;

    if (a > b)
        cout << "Largest Number = " << a; else
        cout << "Largest Number = " << b;

    return 0;
}
```

2. Leap Year

```
#include <iostream>
using namespace std;

int main()
{
    int year;

    cout << "Enter year: ";
    cin >> year;

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) cout << year
        << " is a Leap Year";
    else
        cout << year << " is Not a Leap Year";

    return 0;
}
```

3. Prime Number

```
#include <iostream>
using namespace std;
```

```

intmain()
{
    intn,i;

    cout<<"Enteranumber:"; cin
    >> n;

    for(i=2;i<n;i++)
    {
        if(n%i==0) break;
    }

    if(i==n)
        cout<<"PrimeNumber"; else
        cout<<"NotaPrimeNumber";

    return0;
}

```

4. SwapTwoNumbers

```

#include <iostream>
using namespace std;

intmain(){
    inta=10,b=20,temp;

    temp=a;
    a=b;
    b=temp;

    cout<<"A="<<a<<endl; cout
    <<"B = "<< b;

    return0;
}

```

>BasicProgram C++

```
#include <iostream>
using namespace std;

int main() {
    int a=10;
    int b=20;

    int sum=a+b;

    cout<<"Value of a="<<a<<endl;
    cout<<"Value of b="<<b<<endl;   cout
    <<"Sum = "<< sum << endl;

    return 0;
}
```

>Odd&Even Number

```
#include <iostream>
using namespace std;

int main(){
    int num;

    cout<<"Enter a number:"; cin
    >> num;

    if(num%2==0){
        cout<<"Even Number";
    }else{
        cout<<"Odd Number";
    }

    return 0;
}
```



— Faculty of Computer Science & Applications —
Gokul College of Computer Science & Applications



University Campus, State Highway-41, Sidhpur - 384151, Dist. Patan, Gujarat, INDIA
E: dean.fac.compsci@gokuluniversity.ac.in W: www.gokuluniversity.ac.in M: +91 95124 00808